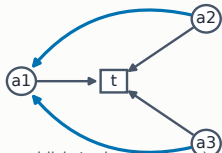


1. Derive dependencies



publish / subscribe (grey)

derived DEPENDS_ON (blue)

$\text{InDeg}(a1) = 2 = \text{afferent coupling}$

2. Count vs. learning

InDeg (count dependents)



GNN on dependency graph



QoS-weighted centrality



0.0 0.5 1.0

Spearman ρ , 12 synthetic architectures

3. Beyond one simulator

No learned engine beats the count on any simulator.

Queue-flow: count keeps partial ρ 0.26 beyond reachability.

Multi-criteria: centrality ranks higher (0.70 vs 0.65).

80% of the critical set needs the top 45% by count.

Count: ms. One simulation: seconds to minutes.